

Hardware — probe, AFE, power, enclosure

Owner: *TBD*

Deliverable for M2: a validated analog front end plus a characterisation dataset.

You are the critical path. PCB fabrication and shipping is 12–18 days including customs, and it cannot be compressed by working harder. Every other track can absorb a slipped week; yours cannot. Order early and order two revisions' worth of components.

1. Safety first — read this before touching anything

You are working near mains potential. Non-negotiable:

- All bench work on the energised side goes through an **isolation transformer** and an **RCD**.
- Use a variac to bring voltage up from zero; never plug a new setup straight into the wall.
- One person never works alone on an energised bench.

- The probe is **non-contact by design**. If at any point your test setup requires galvanic contact with a live conductor, stop and redesign the test.
- The scaled demo rig for the finale must be low voltage (≤ 50 V) with a visible physical break mechanism. Do not bring 230 V onto a hackathon stage.

Document your safety setup with photos. Judges from a utility take this seriously, and showing that you do separates you from teams who clearly did not.

2. Sensing principle

An energised conductor at 230 V creates a 50 Hz electric field in the surrounding air. A conductive plate placed in that field develops a small displacement current through its stray capacitance to the line — typically sub-picofarad at a few tens of centimetres. Measuring that current gives a signal proportional to line voltage, with no CT, no line tap and no outage required to install.

The engineering problem is entirely one of **impedance and shielding**. The source impedance is enormous, so the first stage must be a very high input impedance buffer placed as close to the plate as physically possible.

3. Analog front end

```
[ sense plate ]  
  | (guard ring around plate, driven  
  at buffer output potential)  
  ▼  
  JFET / CMOS opamp buffer      TL072,  
  OPA2140, or a discrete JFET source follower  
  Rin > 1 GΩ, bias resistor 1 GΩ to ground  
  ▼  
  Band-pass 40-60 Hz           2nd order  
  Sallen-Key, rejects DC drift and RF  
  ▼  
  Gain stage, switchable       ×10 / ×100,  
  set at commissioning per standoff  
  ▼  
  Level shift to 0-3.3 V  
  ▼  
  MCU ADC
```

Design notes that matter:

- **Guard ring, driven, not grounded.** A grounded guard shunts your signal. Drive it from the buffer output so the guard sits at the same potential as the plate and stray capacitance to it carries no current.
- **Keep the plate-to-buffer trace under 10 mm.** Put the buffer directly behind the plate on the PCB.
- Use a **PTFE standoff** or guarded PCB layout for the bias resistor; surface leakage across FR4 at 1

GΩ scale is real, especially in Kerala humidity.
Consider a conformal coating.

- Provide a **calibration input** — a small known-amplitude 50 Hz injection so the node can self-test daily and report AFE health.

4. Characterisation — this is your deliverable, not the PCB

The PCB is a means. The dataset is what makes the project credible. For a fixed test conductor, sweep and log:

Variable	Range	Why
Standoff distance	10–80 cm	determines mounting spec and gain setting
Angular position	0–90° off perpendicular	mounting tolerance
Temperature	ambient to 55 °C	enclosure heating in Kerala sun
Relative humidity	40–95%	monsoon; expect leakage effects

Variable	Range	Why
Simulated rain	spray bottle on enclosure	quantify the false-positive scenario
Adjacent conductor	second energised line 1 m away	crosstalk, three-phase reality
Line voltage	50–250 V via variac	linearity

Output: a CSV in `hardware/data/`, plus plots. Then **hand this to the simulator owner** so the synthetic signal model is re-fitted to real probe behaviour instead of a guess.

The rain and humidity rows are the ones a KSEBL engineer will ask about. Have the numbers.

5. Power

Component	Spec
Panel	6 V, 1 W polycrystalline
Cell	1 × 18650 LiFePO ₄ , 1500 mAh

Component	Spec
Charger	CN3791 MPPT or TP5000 for LiFePO4 chemistry
Regulator	low-Iq LDO, quiescent < 10 μ A
Budget	5 days autonomy in monsoon overcast

LiFePO4 over Li-ion: it survives the temperature range on a pole and does not vent violently. Say so in the PPT — it reads as field awareness.

6. BOM target

Under ₹1,800 per node at prototype quantity. Maintain `hardware/BOM.csv` with live pricing and a projected 1,000-unit cost. A cost-effectiveness claim in the problem statement means the panel will ask for the number.

7. Task list

Phase 1 — breadboard (start today)

- [] Source components; order extras of everything
- [] Build buffer + band-pass + gain on breadboard

- ☐ Safe bench setup: isolation transformer, variac, RCD, documented with photos
- ☐ First signal: scope the output with a conductor energised at 230 V, 30 cm standoff
- ☐ Confirm output collapses when the conductor is de-energised

Phase 2 — characterisation → M2

- ☐ Run the full sweep in §4, log to `data/`
- ☐ Plot standoff vs. output, temperature drift, humidity effect
- ☐ Rain test, quantified
- ☐ Hand the dataset to `simulator/` for signal-model re-fit

Phase 3 — PCB

- ☐ KiCad schematic, reviewed by a second person before ordering
- ☐ 2-layer layout: guard ring geometry, short plate trace, guarded bias resistor
- ☐ DRC clean, JLCPCB fab + assembly order placed
- ☐ Order **two** revisions' worth of parts

Phase 4 — bring-up

- ☐ Power rails, quiescent current measurement

- ☐ AFE output matches breadboard characterisation
- ☐ Solar charge cycle verified over a full day
- ☐ Measured average current < 5 mA with firmware running

Phase 5 — mechanical

- ☐ IP65 off-the-shelf enclosure, no drilled holes on the top face
- ☐ 3D-printed pole clamp, fits 90–140 mm pole diameter
- ☐ Plate orientation jig so installers cannot mount it wrong
- ☐ Scaled demo rig: mock pole span, low-voltage conductor, physical cut mechanism

8. Acceptance criteria for M2

1. Repeatable output vs. standoff curve, three runs agreeing within 5%.
2. Signal collapses below 5% of nominal within 200 ms of de-energisation.
3. Rain spray produces a deviation smaller than the 60% collapse threshold, documented.
4. Temperature drift over the full range is slower than the baseline EWMA can track — verified against `firmware/baseline.c`.

9. Gotchas

- **Humidity leakage will surprise you.** At 1 GΩ, a fingerprint across the bias resistor pad changes the reading. Clean with IPA, use a guard trace, conformal coat.
- **Everything is an antenna at this impedance.** Expect the breadboard to behave differently from the PCB. Do not tune thresholds on breadboard data.
- **Three-phase crosstalk is not a corner case.** Real LV distribution has multiple conductors on the same pole. Characterise it now, not at the finale.
- Order the PCB before the design feels finished. Revision two is cheaper than a missed deadline.